

BRIO: Bringing Order to Abstractive Summarization

딥러닝 논문읽기 모임

자연어처리팀 : 조해창, 김유진, 변현정, 박산희, 이기성(발표자)

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1. Background

ROUGE

- N-gram을 기준으로 Recall, Precision, F1을 계산

Ex)

Reference Text : 딥러닝 논문 읽기 모임은 유익하다.

Inference Text : 논문 모임은 매우 유익하다.

- Recall : 0.6
- Precision : 0.75
- F1 : 0.67

Exposure bias

- Train 과정에서는 Teacher forcing이 적용
- Inference 과정에서는 이전 스텝에서 예측한 토큰을 사용

Contrastive Learning

- 대상들의 차이를 명확하게 보여줄 수 있도록 학습하는 것
- Metric Learning의 일종

$$\text{loss}(x_1, x_2, y) = \max(0, -y * (x_1 - x_2) + \text{margin})$$

2. Abstract

- 추상 요약은 Maximum likelihood estimation으로 학습(deterministic – one point)
- Non-deterministic 분포를 가정하는 새로운 학습 paradigm을 제시
- 서로 다른 생성 요약 후보에 확률 질량을 부여(품질에 따라서)

3. Introduction

- Autoregressive Manner → Exposure bias
- Metric(Rouge)와 실제 생성 요약의 확률 간의 불일치를 해결하는 것이 목적

Reference samples

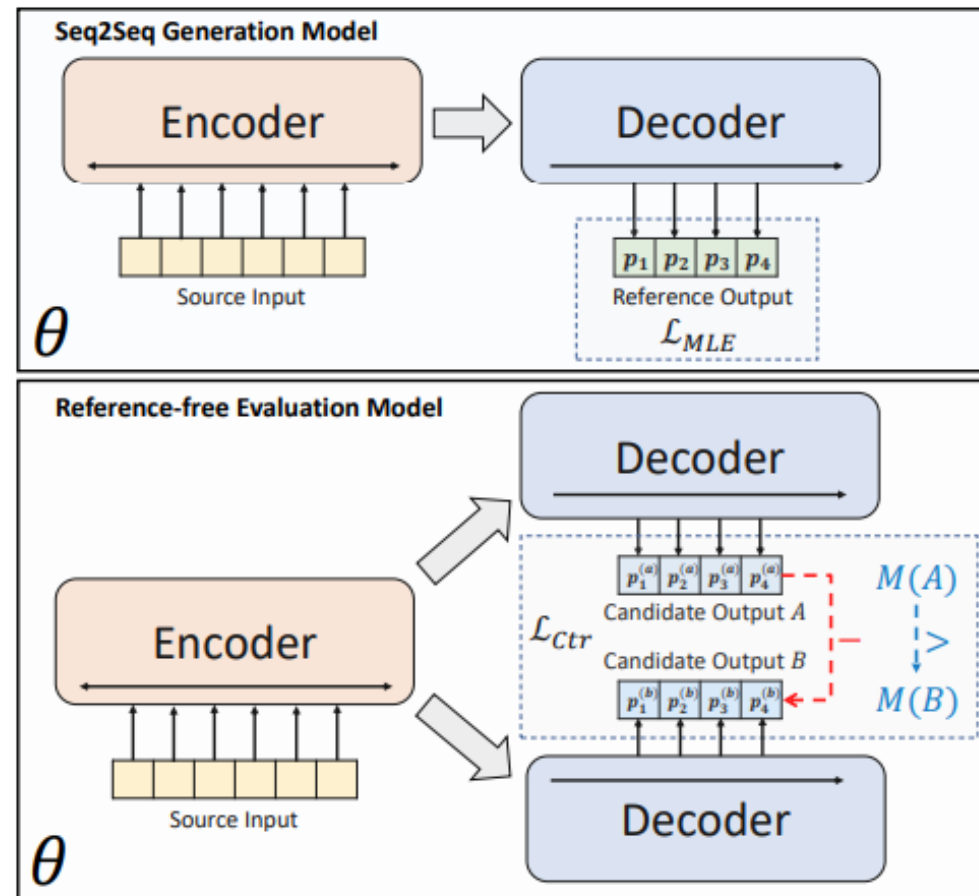
- 딥러닝 논문 읽기 모임은 유익하다.

Generate samples

- 딥러닝 논문 읽기 모임은 유익하다. / P : 0.5
- 논문 모임은 매우 유익하다. / P : 0.7

System	R-1	R-2	R-L	Acc.(%)
High	53.99	29.85	51.12	100.00
Low	33.48	10.85	30.45	0.00
BART	44.88	21.68	41.92	54.80
Ours	50.10	26.29	47.19	79.63

- 모델에 두 개의 역할을 부여 → generation & Evaluation
- Generation Model : MLE Loss로 학습
- Evaluation Model : Contrastive Loss로 학습
- **Main contribution** : 모델이 학습할 타겟 분포를 deterministic 한 분포에서 non-deterministic 분포로 변경



4. Neural Abstractive Summarization

Training Objective

- D : Source document
- S : Reference summary / s : summary token
- g : Function

$$S \leftarrow g(D) \quad (1)$$

$$\theta^* = \operatorname{argmax}_{\theta} \sum_i \log p_{g_{\theta}}(S^{*(i)} | D^{(i)}; \theta) \quad (2)$$

$$\begin{aligned} \mathcal{L}_{xent} = \\ - \sum_{j=1}^l \sum_s p_{\text{true}}(s | D, S_{<j}^*) \log p_{g_{\theta}}(s | D, S_{<j}^*; \theta) \end{aligned} \quad (3)$$

$$p_{\text{true}}(s | D, S_{<j}^*) = \begin{cases} 1 & s = s_j^* \\ 0 & s \neq s_j^* \end{cases} \quad (4)$$

$$p_{\text{true}}(s | D, S_{<j}^*) = \begin{cases} 1 - \beta & s = s_j^* \\ \frac{\beta}{N-1} & s \neq s_j^* \end{cases} \quad (5)$$

Inference and Exposure Bias

$$p_{g\theta}(s_t|D, S_{<t}; \theta) \quad (6)$$

5. Coordination Abstractive Models

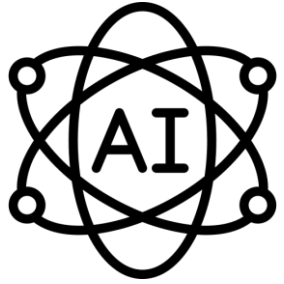
Contrastive Learning for Coordination

$$\begin{cases} p_{\text{true}\dagger}(S|D) = 1 - \beta & S = S^* \\ \sum_{S \in \mathcal{S}} p_{\text{true}\dagger}(S|D) = \beta & S \neq S^* \\ p_{\text{true}\dagger}(S_i|D) > p_{\text{true}\dagger}(S_j|D) & \forall S_i, S_j \in \hat{\mathcal{S}}, \\ & M(S_i) > M(S_j) \end{cases} \quad (7)$$

$$f(S) = \frac{\sum_{t=1}^l \log p_{g\theta}(s_t|D, S_{<t}; \theta)}{|S|^\alpha} \quad (9)$$

$$\mathcal{L}_{ctr} = \sum_i \sum_{j>i} \max(0, f(S_j) - f(S_i) + \lambda_{ij}) \quad (8)$$

$$\mathcal{L}_{mul} = \mathcal{L}_{xent} + \gamma \mathcal{L}_{ctr} \quad (10)$$



Reference samples

- 딥러닝 논문 읽기 모임은 유익하고 재미있다.



MLE Loss



Generate samples

- 딥러닝 모임은 유익
- 논문 모임은 매우 유익하다.
- 딥러닝 논문 읽기 모임은 재미있다.
- 모임은 재미있다.

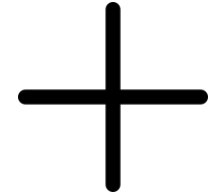


Generate samples(Sorted by ROUGE)

1. 딥러닝 논문 읽기 모임은 재미있다.
2. 논문 모임은 매우 유익하다.
3. 딥러닝 모임은 유익
4. 모임은 재미있다.



Contrastive Loss



6. Experiments

Experimental Settings / Implementation Details

- Datasets : CNNDM, Xsum, NYT
- Baselines : BART, PEGASUS, Gsum, SimCLS, GOLD, SeqCo, ConSum
- BRIO-Ctr : Contrastive loss only
- BRIO-Mul : CE loss + Contrastive loss

Results

System	R-1	R-2	R-L
CNNDM			
BART*	44.16	21.28	40.90
PEGASUS*	44.17	21.47	41.11
GSum*	45.94	22.32	42.48
ConSum*	44.53	21.54	41.57
SeqCo*	45.02	21.80	41.75
GOLD- p *	45.40	22.01	42.25
GOLD- s *	44.82	22.09	41.81
SimCLS*	46.67	22.15	43.54
BART [‡]	44.29	21.17	41.09
BRIO-Ctr	47.28 [†]	22.93 [†]	44.15 [†]
BRIO-Mul	47.78[†]	23.55[†]	44.57[†]
XSum			
BART*	45.14	22.27	37.25
PEGASUS*	47.21	24.56	39.25
GSum*	45.40	21.89	36.67
ConSum*	47.34	24.67	39.40
SeqCo*	45.65	22.41	37.04
GOLD- p *	45.75	22.26	37.30
GOLD- s *	45.85	22.58	37.65
SimCLS*	47.61	24.57	39.44
PEGASUS [‡]	47.46	24.69	39.53
BRIO-Ctr	48.13 [†]	25.13 [†]	39.84 [†]
BRIO-Mul	49.07[†]	25.59[†]	40.40[†]
NYT			
BART [‡]	55.78	36.61	52.60
BRIO-Ctr	55.98	36.54	52.51
BRIO-Mul	57.75[†]	38.64[†]	54.54[†]

Analysis

- Coefficients of the Multi-Task Loss

$$\mathcal{L}_{mul} = \mathcal{L}_{xent} + \gamma \mathcal{L}_{ctr} \quad (10)$$

Coefficient (γ)	R-1	R-2	R-L
0 (BART)	44.29	21.17	41.09
0.1	45.08	21.63	41.71
1	46.01	22.22	42.68
2	46.36	22.79	43.07
5	46.91	23.03	43.63
10	47.22	23.31	43.94
100	47.78	23.55	44.57
1000	46.83	22.17	43.68
$+\infty$ (BRIO-Ctr)	47.28	22.93	44.15

Analysis

- Generation-Finetuning as a Loop

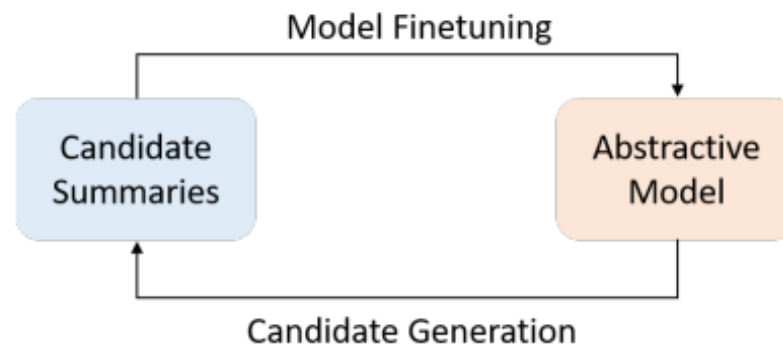


Figure 2: Loop of candidate generation and model finetuning.

System	R-1	R-2	R-L
BART	44.29	21.17	41.09
BRIO-Mul	47.78	23.55	44.57
BRIO-Loop	48.01[†]	23.80[†]	44.67[†]

Table 4: Results on CNNDM when the pre-trained model are fine-tuned twice. **BRIO-Loop** is trained on the candidates generated by **BRIO-Mul**. [†]: significantly better than the baseline (BART) ($p < 0.01$). R-1/2/L are ROUGE-1/2/L F₁ scores.

Analysis

- Increasing the Beam Width

Beams	BART		BRIO-Mul	
	R-1	R-2	R-1	R-2
4	44.29	21.17	47.78	23.55
10	43.83	20.76	47.98	23.81
20	43.53	20.49	48.07	23.92
50	43.06	20.05	48.18	24.01
100	42.79	19.76	48.23	24.09

Table 5: Results on CNNDM with different beam widths (the number of beams) used in beam search. The default beam width is 4. R-1/2 are the ROUGE-1/2 F_1 scores.

Analysis

- Training with Different Evaluation Metrics

System	R-1	R-2	R-L	BS
BART	44.29	21.17	41.09	27.38
BRIO-Mul (R)	47.78	23.55	44.57	32.11
BRIO-Mul (B)	47.53	23.22	44.37	32.59

Table 6: Results on CNNDM using different evaluation metrics as M in Eq.7. BRIO-Mul (R) is trained with candidate summaries ordered by ROUGE scores, while BRIO-Mul (B) is trained with candidate summaries ordered by BERTScore. R-1/2/L are ROUGE-1/2/L F_1 scores. BS denotes BERTScore.

Analysis

- Novel n-grams

$$\text{Novelty}(D, S^*) = \frac{\sum_{g \in G_{S^*}} \mathbb{1}(g \notin G_D)}{|G_{S^*}|} \quad (11)$$

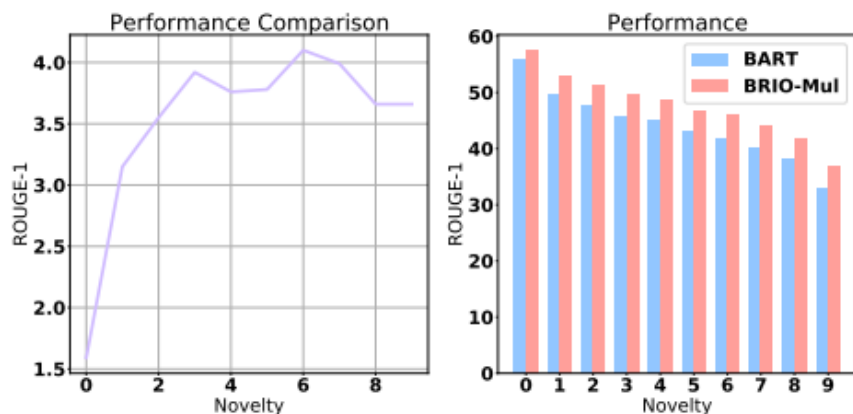


Figure 3: Performance comparison (BART v.s. BRIO-Mul) w.r.t. reference summary novelty. The x-axis represents different buckets of test examples grouped by reference summary novelty (Eq. 11). Larger x-coordinates correspond to examples of which the reference summaries have higher novelty. The left figure shows the performance improvement of our model compared with the baseline model, while the right one shows model performance.

System	Unigram	Bigram
Reference	.1110	.4865
BART	.0101	.0924
BRIO-Mul	.0262	.2381

Table 7: Ratio of novel n -grams of different models on CNNDM. Novel n -grams are those that appear in the summaries but not in the source documents.

Analysis

- Rank Correlation

	Own	PEGASUS
BART	.0470	.1205
BRIO-Mul	.1839[†]	.2768[†]

Table 8: Rank Correlation between the model’s estimated probabilities of the candidate summaries and the quality scores (ROUGE) of the candidate summaries on CNNDM. **Own** stands for the candidates generated by the models themselves, while **PEGASUS** stands for the candidates generated by the pre-trained PEGASUS model. [†]: significantly better than the baseline model (BART) ($p < 0.01$).

$$f(S) = \frac{\sum_{t=1}^l \log p_{g_\theta}(s_t | D, S_{<t}; \theta)}{|S|^\alpha} \quad (9)$$

Token-level Calibration

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (12)$$

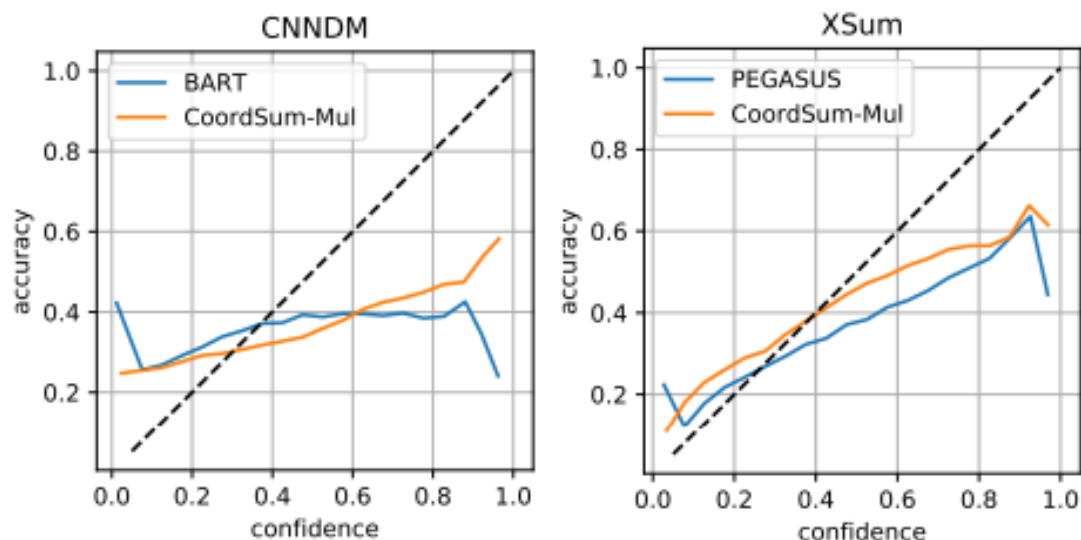


Figure 4: Reliability graphs on the CNNDM and XSum datasets. The accuracy of model’s predictions is plotted against the model’s confidence on these predictions.

Dataset	System	ECE	Acc	Conf
CNNDM	BART	.4097	.3711	.7365
	BRIO-Mul	.2719	.4271	.6652
XSum	PEGASUS	.2369	.4688	.6990
	BRIO-Mul	.1423	.4744	.5881

Table 9: Expected Calibration Error (ECE), accuracy (Acc) and confidence (Conf) on the test set of CNNDM and XSum.

Few-shot Fine-tuning

Dataset	System	R-1	R-2	R-L
CNNDM	BART	44.29	21.17	41.09
	BRIO-Few	45.81	21.91	42.61
XSum	PEGASUS	47.46	24.69	39.53
	BRIO-Few	47.95	24.89	39.71

Table 11: Few-shot Fine-tuning. **BRIO-Few** is trained on only 100/1000 training examples on CNNDM and XSum respectively. R-1/2/L are ROUGE-1/2/L F_1 scores.

Case Study on CNNDM

System	Summary
Reference	chelsea forward tammy abraham nets first-half double for chelsea. dominic solanke adds a third late on as chelsea look set to win trophy. manchester city struggle without injured star thierry ambrose. read: mourinho warns his young chelsea players he can not play them all. click here to read our match report from man city 's academy stadium.
BART	tammy abraham scored twice in the first half to give chelsea the lead. isaac buckley-ricketts levelled the game for manchester city. dominic solanke scored late on to put a gloss on the scoreline. click here to read sportsmail's player ratings from the youth cup final.
BRIO-Mul	chelsea beat manchester city 3-1 in the youth cup final at the etihad stadium. tammy abraham scored twice in the first half to give chelsea the lead. dominic solanke scored late on to seal the win for the home side.
Reference	alejandro valverde won ahead of julian alaphilippe and michael albasini. chris froome finished 123rd after a crash during the final 12 kilometres. team sky's sports director gabriel rasch praised froome for finishing. rasch said froome was 'banged up' but expects to ride tour de romandie.
BART	movistar rider alejandro valverde won fleche wallonne on wednesday. team sky's chris froome fell in the final 12km but finished the race. philippe gilbert pulled out of the race after a bad crash 50km from the end. click here for more cycling news.
BRIO-Mul	alejandro valverde defended his fleche wallonne title in belgium on wednesday. movistar rider finished ahead of julian alaphilippe and michael albasini. team sky's chris froome fell in the final 12km of the race but finished in 123rd. froome was involved in a crash but finished the race despite being 'banged up'
Reference	manuel pellegrini won the premier league and capital one cup last season. city currently sit fourth in the league table - 12 points behind chelsea. pellegrini's contract expires at the end of the 2015-16 season. city players have been impressed with vieira's work with the youth team. pep guardiola is city's first-choice to succeed pellegrini at the etihad.
BART	manuel pellegrini's future at manchester city is under scrutiny. patrick vieira is highly-respected among the city players. city's first-choice managerial option is bayern munich boss pep guardiola. click here for all the latest manchester city news. click here for more premier league news.
BRIO-Mul	manchester city players have backed patrick vieira to replace manuel pellegrini as manager of the club. the frenchman is highly-respected among the players at the etihad stadium. pellegrini's future at the club is under scrutiny after a disappointing season. city's first-choice manager is current bayern munich boss pep guardiola.

Table 10: Case Study on CNNDM. BRIO-Mul learns to ignore the noise pattern ("click here") while BART cannot.

6. Conclusion And Future Work

- Metric에 따른 점수와 candidate outputs의 확률을 활용한 Contrastive한 학습 방법을 제시
- 추상 요약 뿐만 아니라 기계 번역 Task에도 확장 가능성
- 강화 학습에 적용 가능성이 존재
- Diverse beam search 외의 다른 생성 방법을 활용한 성능 개선 가능성